**Experiment No 04**

**Aim:**

To develop a blockchain-powered web application using Solidity on Remix IDE, connect it with MetaMask wallet, and deploy a smart contract on the Sepolia Testnet to record and retrieve student marks.

**Theory:**

**Blockchain Technology**

A blockchain is a distributed, immutable ledger maintained across a network of nodes rather than a single central server. Each block contains a set of transactions, a timestamp, and a cryptographic hash of the previous block, forming a tamper-evident chain. Once data is written to a block and confirmed by the network's consensus mechanism, it cannot be altered or deleted, which is what makes blockchains trustworthy for recording transactions without relying on a central authority.

**Smart Contracts**

A smart contract is self-executing program code deployed directly onto a blockchain. Once deployed, its logic runs exactly as written, and its execution cannot be changed, censored, or reversed by any single party — not even the contract's creator, unless the code explicitly allows it (e.g., via an owner-only function). Smart contracts have their own address on the blockchain, can store data permanently in "state variables," and expose functions that anyone can call to read or modify that data. Every function call that changes data is itself a transaction, requiring computational cost ("gas") and a cryptographic signature from the caller's wallet.

**Solidity**

Solidity is a statically-typed, contract-oriented programming language designed specifically for writing smart contracts on Ethereum-compatible blockchains. It supports familiar programming constructs — structs, mappings, modifiers, events, and access control — while enforcing rules unique to blockchain environments, such as explicit visibility for functions (public, private) and built-in overflow/underflow protection (from Solidity 0.8 onward). Contracts written in Solidity are compiled into EVM (Ethereum Virtual Machine) bytecode before deployment.

**Remix IDE**

Remix is a free, browser-based Integrated Development Environment for writing, compiling, debugging, testing, and deploying Solidity smart contracts, with no local installation required. It provides a file explorer, an in-browser Solidity compiler, and a "Deploy & Run Transactions" module that can connect to different environments — a fully simulated in-browser blockchain (Remix VM), or a real network via an injected wallet provider like MetaMask.

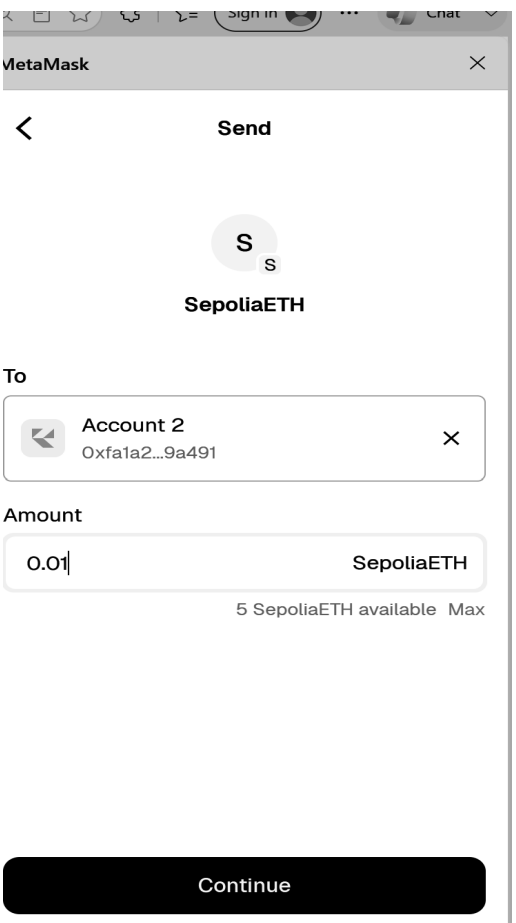
**MetaMask Wallet**

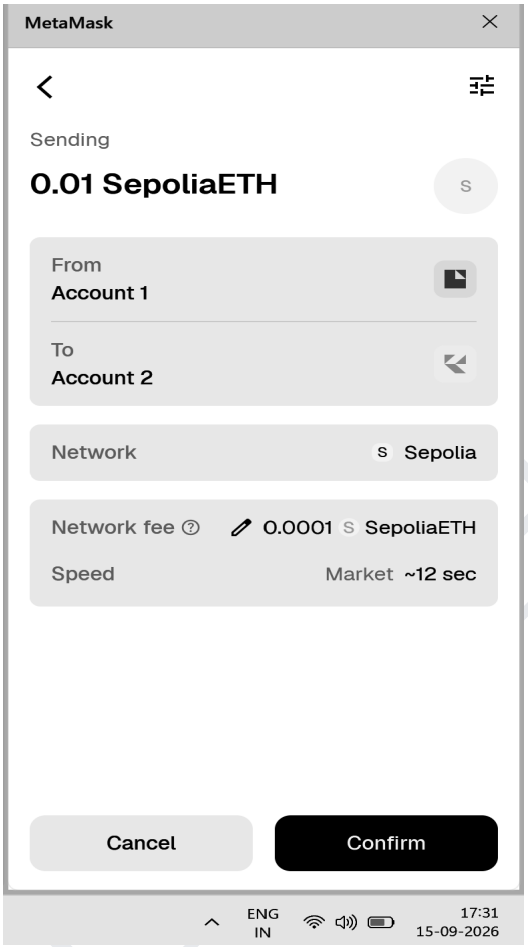
MetaMask is a cryptocurrency wallet available as a browser extension that manages a user's private keys locally and never exposes them to the websites it interacts with. It acts as an intermediary between a dApp (like Remix) and the blockchain: whenever a transaction needs to be sent — whether deploying a contract or calling a function that modifies data — MetaMask intercepts the request, displays its details (network, gas fee, destination address), and requires explicit user confirmation before signing and broadcasting it. This "injected provider" model is what lets browser-based tools like Remix send real transactions without ever handling the user's private key directly.

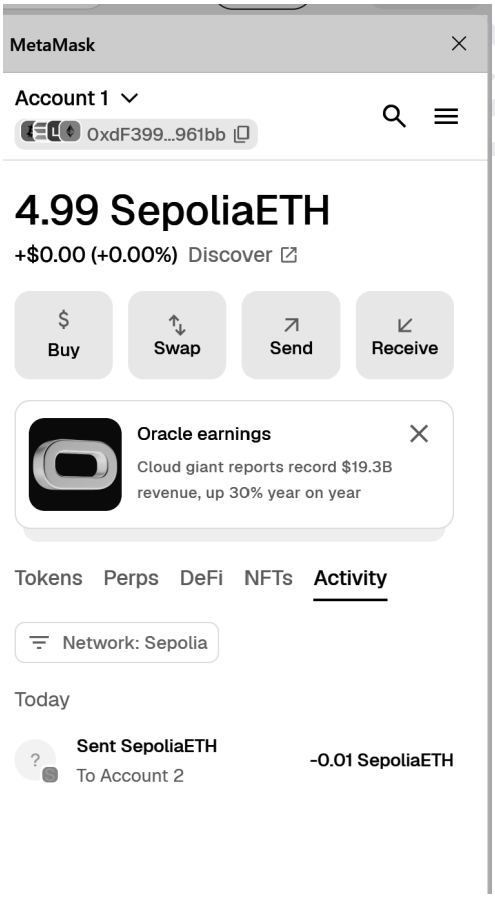
**Testnets and Gas**

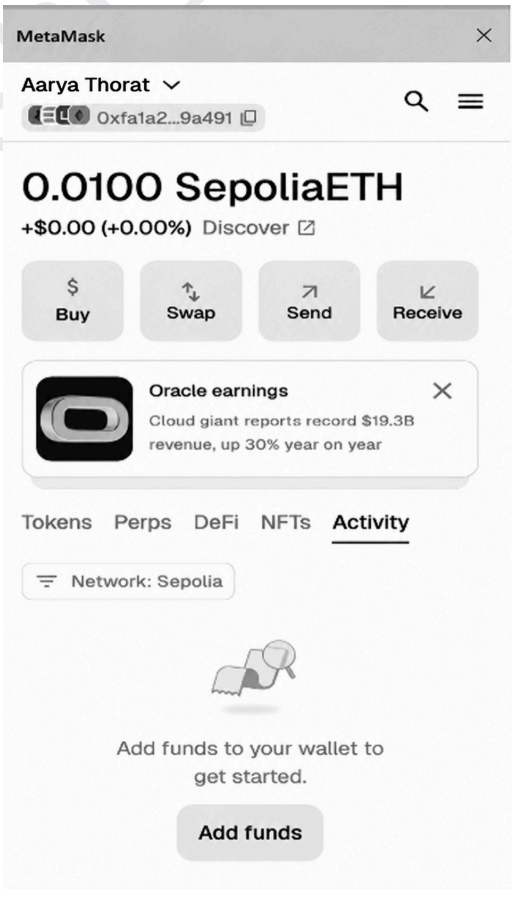
A testnet (such as Sepolia) is a separate blockchain network that mirrors the behavior, rules, and tooling of the Ethereum mainnet but uses worthless test tokens instead of real cryptocurrency. Developers obtain free test ETH from "faucets" to pay for gas — the computational fee required to execute transactions and store data on-chain — allowing contracts to be built, deployed, and tested realistically without financial risk. Once a contract is confirmed to work correctly on a testnet, the same code can be deployed to the mainnet with real funds.

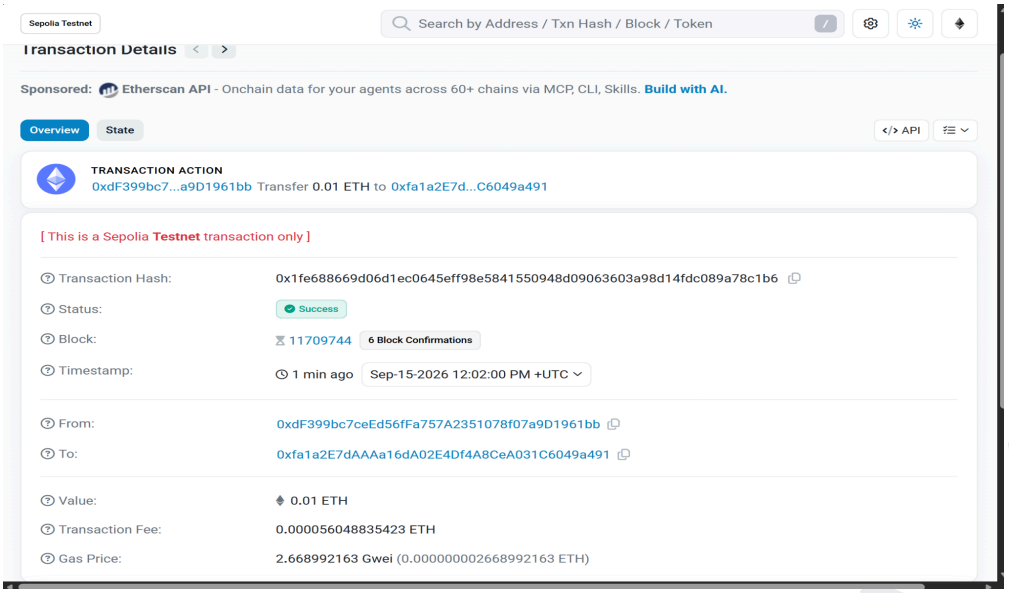
Vedant_Joshi_D20A_46

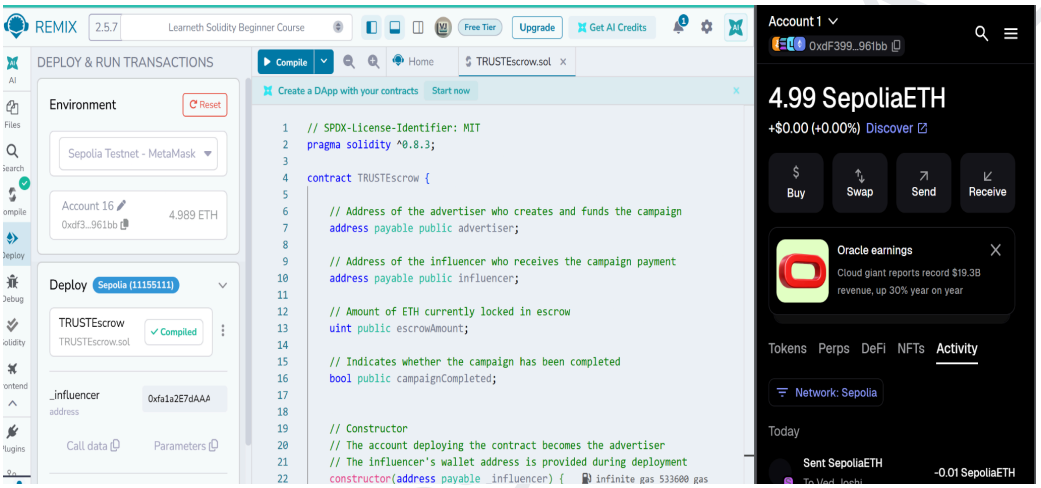

MetaMask
Send
S
SepoliaETH
To
Account 2
0xfa1a2...9a491
Amount
0.01
SepoliaETH
5 SepoliaETH available Max
Continue

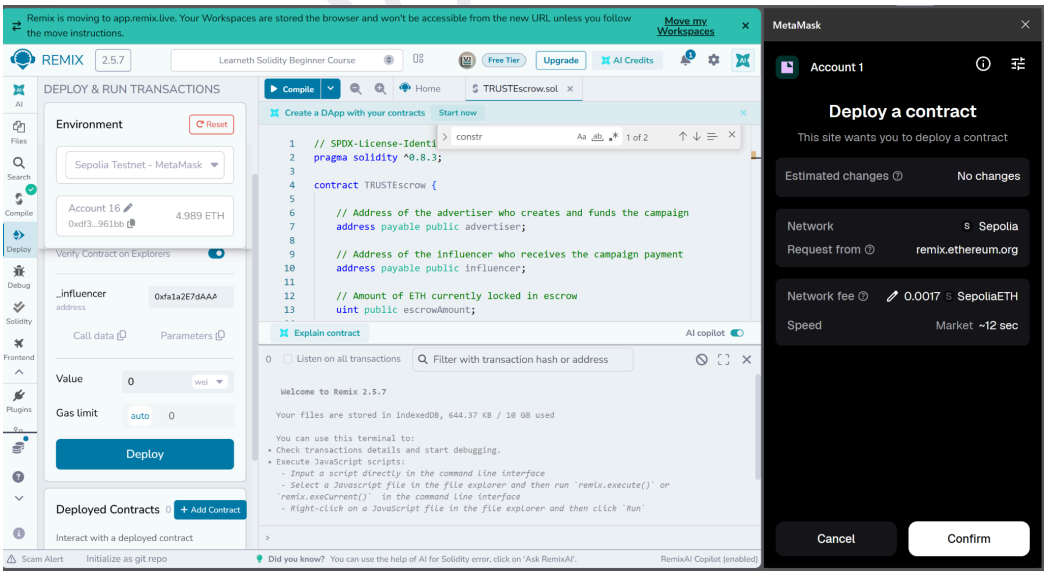

MetaMask
Sending
0.01 SepoliaETH
From
Account 1
To
Account 2
Network
Sepolia
Network fee
0.0001 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm
ENG
IN
17:31
15-09-2026

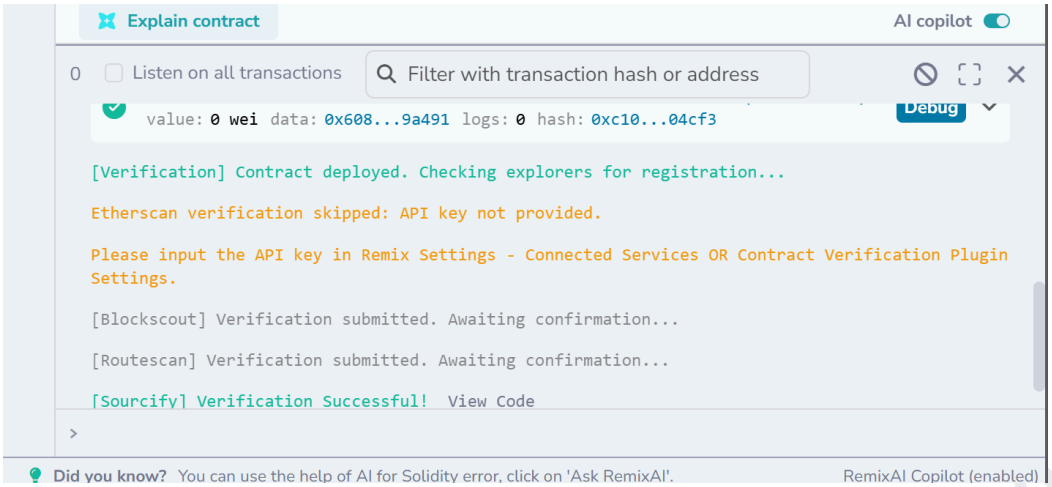

MetaMask
Account 1
0xdF399...961bb
4.99 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Oracle earnings
Cloud giant reports record $19.3B revenue, up 30% year on year
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
Today
Sent SepoliaETH
To Account 2
-0.01 SepoliaETH

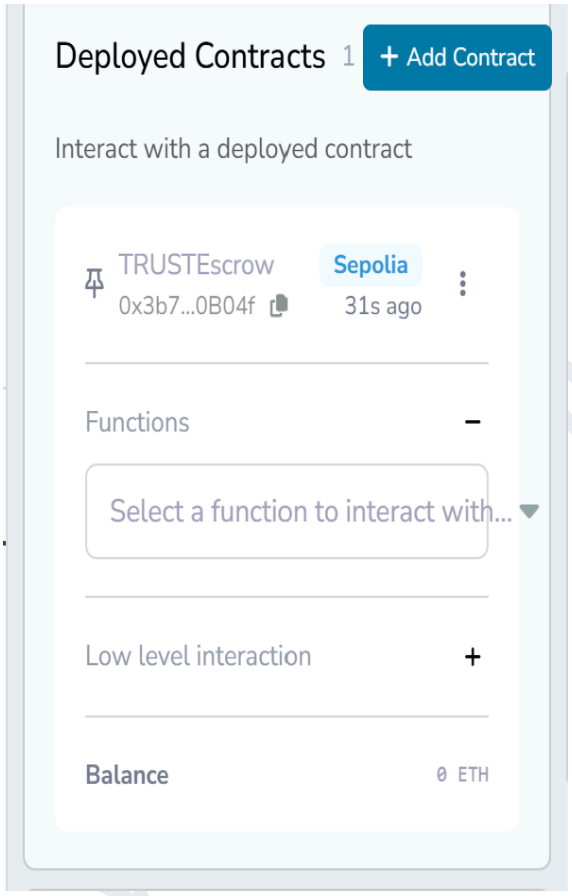

MetaMask
Aarya Thorat
0xfa1a2...9a491
0.0100 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Oracle earnings
Cloud giant reports record $19.3B revenue, up 30% year on year
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
Add funds to your wallet to get started.
Add funds

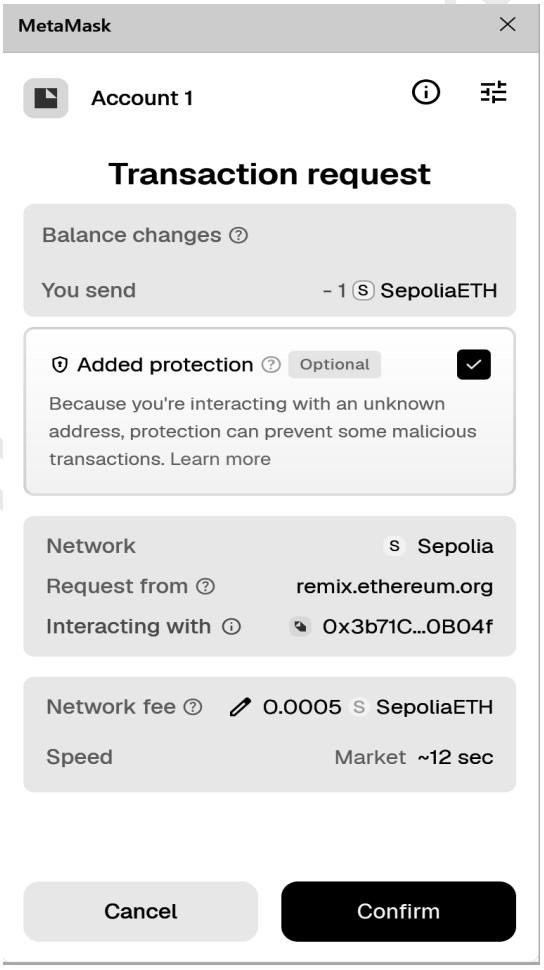

Sepolia Testnet
Search by Address / Txn Hash / Block / Token
Transaction Details
Sponsored: Etherscan API - Onchain data for your agents across 60+ chains via MCP, CLI, Skills. Build with AI.
Overview
State
</> API
TRANSACTION ACTION
0xdF399bc7...a9D1961bb Transfer 0.01 ETH to 0xfa1a2E7d...C6049a491
[ This is a Sepolia Testnet transaction only ]
Transaction Hash:
0x1fe688669d06d1ec0645eff98e5841550948d09063603a98d14fdc089a78c1b6
Status:
Success
Block:
11709744
6 Block Confirmations
Timestamp:
1 min ago
Sep-15-2026 12:02:00 PM +UTC
From:
0xdF399bc7ceEd56fFa757A2351078f07a9D1961bb
To:
0xfa1a2E7dAAAa16dA02E4Df4A8CeA031C6049a491
Value:
0.01 ETH
Transaction Fee:
0.000056048835423 ETH
Gas Price:
2.668992163 Gwei (0.000000002668992163 ETH)

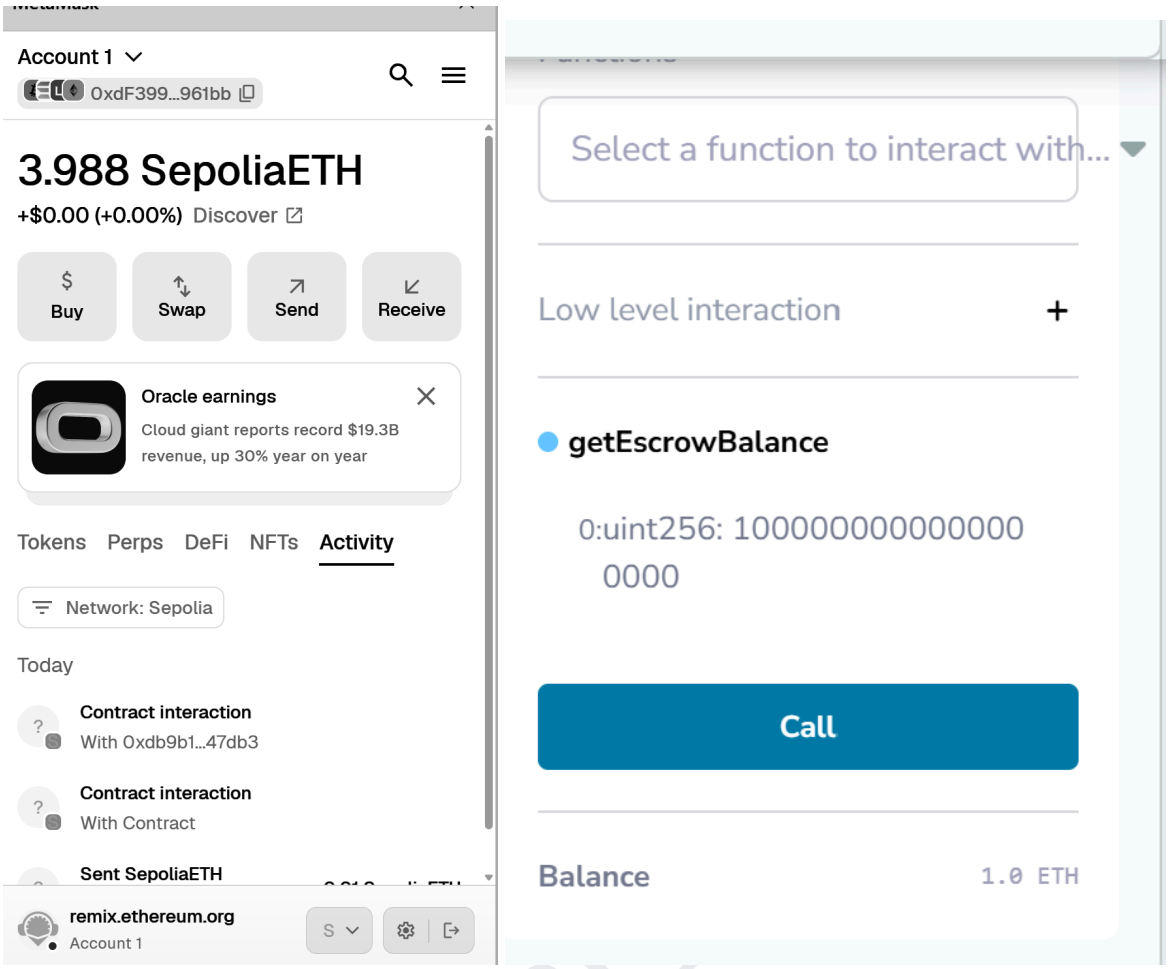

REMIX
2.5.7
Learneth Solidity Beginner Course
Free Tier
Upgrade
Get AI Credits
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Sepolia Testnet - MetaMask
Account 16
0xdf3...961bb
4.989 ETH
Deploy
Sepolia (11155111)
TRUSTEscrow
TRUSTEscrow.sol
Compiled
_influencer
address
0xfa1a2E7dAAA
Call data
Parameters
Compile
Home
TRUSTEscrow.sol
Create a DApp with your contracts
Start now
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.3;
contract TRUSTEscrow {
// Address of the advertiser who creates and funds the campaign
address payable public advertiser;
// Address of the influencer who receives the campaign payment
address payable public influencer;
// Amount of ETH currently locked in escrow
uint public escrowAmount;
// Indicates whether the campaign has been completed
bool public campaignCompleted;
// Constructor
// The account deploying the contract becomes the advertiser
// The influencer's wallet address is provided during deployment
constructor(address payable _influencer) {
infinite gas 533600 gas
Account 1
0xdF399...961bb
4.99 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Oracle earnings
Cloud giant reports record $19.3B revenue, up 30% year on year
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
Today
Sent SepoliaETH
-0.01 SepoliaETH


Remix is moving to app.remix.live. Your Workspaces are stored the browser and won't be accessible from the new URL unless you follow the move instructions.
Move my Workspaces
REMIX
2.5.7
DEPLOY & RUN TRANSACTIONS
Environment
Sepolia Testnet - MetaMask
Account 16
4.989 ETH
Value
0
Gas limit
auto
Deploy
Deployed Contracts
Add Contract
Interact with a deployed contract
Compile
TRUSTEscrow.sol
Create a DApp with your contracts
Start now
// SPDX-License-Identi
pragma solidity ^0.8.3;
contract TRUSTEscrow {
// Address of the advertiser who creates and funds the campaign
address payable public advertiser;
// Address of the influencer who receives the campaign payment
address payable public influencer;
// Amount of ETH currently locked in escrow
uint public escrowAmount;
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or address
Welcome to Remix 2.5.7
Your files are stored in indexedDB, 644.37 KB / 10 GB used
You can use this terminal to:
• Check transactions details and start debugging.
• Execute JavaScript scripts:
- Input a script directly in the command line interface
- Select a Javascript file in the file explorer and then run `remix.execute()` or `remix.exeCurrent()` in the command line interface
- Right-click on a JavaScript file in the file explorer and then click `Run`
Scam Alert
Initialize as git repo
Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'.
RemixAI Copilot (enabled)
MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
remix.ethereum.org
Network fee
0.0017 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


Explain contract
AI copilot
0
Listen on all transactions
Filter with transaction hash or address
value: 0 wei data: 0x608...9a491 logs: 0 hash: 0xc10...04cf3
Debug
[Verification] Contract deployed. Checking explorers for registration...
Etherscan verification skipped: API key not provided.
Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.
[Blockscout] Verification submitted. Awaiting confirmation...
[Routescan] Verification submitted. Awaiting confirmation...
[Sourcify] Verification Successful! View Code
>
Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'.
RemixAI Copilot (enabled)


Deployed Contracts 1
+ Add Contract
Interact with a deployed contract
TRUSTEscrow
Sepolia
0x3b7...0B04f
31s ago
Functions
Select a function to interact with...
Low level interaction
Balance
0 ETH


MetaMask
Account 1
Transaction request
Balance changes
You send
- 1 SepoliaETH
Added protection
Optional
Because you're interacting with an unknown address, protection can prevent some malicious transactions. Learn more
Network
Sepolia
Request from
remix.ethereum.org
Interacting with
0x3b71C...0B04f
Network fee
0.0005 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm




**Conclusion -**

The StudentMarks smart contract was successfully compiled, deployed, and verified on the Sepolia Testnet using Remix IDE integrated with MetaMask. Both write operations (addStudent) and read operations (getMarks) were tested and confirmed to work correctly, with the transaction and event logs visible on Etherscan. This experiment demonstrated the complete workflow of wallet-based transaction signing and on-chain data storage and retrieval.

Ved_Joshi